ANALYSTLAB AFRICA

Experience Lab Internship Programme
Official Weekly Assignment
Week 3: Hospital Operations, Cost & Patient Flow Intelligence

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 3
Project	Healthcare Analytics and AI Implementation
Project Title	Hospital Operations, Cost & Patient Flow Intelligence
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 3 (Wednesday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 3 Submission Form

1. Project Background

Client Scenario

A healthcare organisation wants to improve its understanding of hospital operations, encounter costs, payer coverage, procedure utilisation, and patient return visits.

The organisation has provided historical healthcare datasets containing encounter records, patient information, procedure records, payer details, organisation information, and a data dictionary.

As an intern at AnalystLab Africa, you have been assigned to a healthcare intelligence project team. Your responsibility is to use the skills acquired in your learning track to investigate a defined business problem and develop a solution that supports evidence-based operational decision-making.

You are expected to go beyond producing charts, writing code, or building an application. Your work must demonstrate sound reasoning, appropriate methodology, validation, professional documentation, and an understanding of the limitations of the available data.

2. Project Objectives

By the end of this assignment, you should be able to:

1. Translate a healthcare operations challenge into a clearly defined problem.
2. Assess the suitability, structure, quality, and limitations of healthcare datasets.
3. Apply the appropriate analytical or technical methods for your learning track.
4. Validate your findings or solution using suitable evidence.
5. Communicate results and recommendations professionally.
6. Apply responsible data handling, privacy, and documentation practices.
7. Produce a project deliverable that can be reviewed, tested, and understood by another professional.

3. Dataset Resources

Use the healthcare dataset pack provided by AnalystLab Africa.

Resource	Primary purpose
encounters.csv	Analyse encounter volumes, encounter classes, duration, costs, and payer coverage
patients.csv	Support appropriate patient demographic and patient-level analysis
procedures.csv	Analyse procedure frequency, costs, and utilisation
payers.csv	Support payer-related analysis
organizations.csv	Understand the organisation represented in the data
data_dictionary.csv	Understand field definitions and intended relationships
hospital_analytics_questions.sql	Reference SQL questions for healthcare analysis
create_hospital_db.sql	Optional SQL database setup resource

Important: The SQL question file is a reference resource, not a substitute for your own analysis. You are expected to understand your selected problem and explain your approach.

Before beginning, inspect the datasets and confirm that the fields required for your chosen analysis are available. Do not assume that every field described in the data dictionary is present in every CSV file.

4. General Instructions for All Interns

Every intern must complete the following foundational activities before beginning the track-specific tasks.

Task 1: Understand the business problem

Write a brief explanation of the healthcare organisation's challenge and identify the operational question your work will address.

Your explanation must include:

- The problem you intend to investigate or solve.
- Why the problem matters to healthcare operations.
- The intended user of your final deliverable.
- The decision your findings or solution could support.
- The limitations of the available evidence.

Task 2: Inspect the datasets

Review the relevant datasets and document:

- The tables and fields you intend to use.
- The number of records and columns in each selected file.
- Missing values, duplicate records, inconsistent data types, and date issues.
- The relationships between relevant tables.
- Any limitations that could affect the validity of your work.

Do not automatically remove duplicates or replace missing values without first understanding their meaning and potential effect on your analysis.

Task 3: Define success criteria

Identify at least three measurable criteria that will help determine whether your work is successful.

Examples include:

- Analytical calculations reconcile with an independent check.
- A model outperforms an appropriate baseline on a defined evaluation metric.
- An AI application provides supported answers to a test set of questions.
- A project plan contains clear dependencies, milestones, owners, and acceptance criteria.

Select criteria that are appropriate to your track.

Task 4: Maintain professional standards

- Use clear file names and an organised project folder.
- Document assumptions and important decisions.
- Attribute external sources, code, and other materials used.
- Include evidence that your work has been tested or validated.
- Do not expose patient names, addresses, or other unnecessary identifying information in reports, screenshots, dashboards, or public repositories.
- Do not make clinical claims or treatment recommendations unsupported by the supplied data.

Database safety: The provided SQL setup script begins by dropping and recreating a database named `hospital_db`. Inspect the script before running it and use only an isolated practice environment. Never execute it against a production or existing database containing important information.

5. TRACK-SPECIFIC ASSIGNMENTS

TRACK A: DATA ANALYTICS

Assignment Title

Healthcare Operations & Financial Intelligence Dashboard

Objective

Develop an interactive healthcare operations dashboard that helps decision-makers understand encounter patterns, costs, payer coverage, and procedure utilisation.

Task 1: Prepare and model the data

1. Import the relevant healthcare datasets into Power BI or an appropriate analytical tool.
2. Assess data quality and document the issues discovered.
3. Identify the relationships between encounters, patients, procedures, and payers.
4. Create an appropriate data model where the available keys and data support it.
5. Document any transformations performed and explain why they were necessary.

Task 2: Analyse healthcare operations

Investigate at least four of the following questions:

1. How have encounter volumes changed over time?
2. Which encounter classes account for the largest proportion of recorded encounters?
3. What proportion of encounters lasted more than 24 hours, where valid start and end timestamps are available?
4. How do total claim costs vary across encounter classes?
5. How much payer coverage is recorded, and how many encounters have zero recorded payer coverage?
6. Which procedures occur most frequently?
7. Which procedures have the highest average base cost, considering both frequency and the limitations of the data?
8. How do average total claim costs vary by payer?

For each question, state the relevant metric, calculation method, and business interpretation.

Task 3: Build the dashboard

Your dashboard must contain:

- An overview page with key operational indicators.

- A trend analysis of encounter volumes.
- An encounter-class comparison.
- A cost and payer-coverage analysis.
- A procedure-utilisation analysis.
- Appropriate filters and interactive visual elements.

Include at least five meaningful KPIs, provided the necessary data is available. Document the definitions of your KPIs so that another analyst can reproduce the calculations.

Task 4: Validate the dashboard

- Independently verify at least three important calculations.
- Check whether totals change correctly when filters are applied.
- Investigate discrepancies between the dashboard and the source data.
- Document any metric that could be misleading because of missing information or ambiguous definitions.

Task 5: Produce business recommendations

Write at least three evidence-based recommendations.

For each recommendation, state the finding that supports it, the proposed action, the intended benefit, and any limitation or additional information required.

Required Deliverables

1. Power BI `.pbix` file or equivalent working dashboard.
2. Data-quality assessment.
3. KPI definition sheet.
4. Dashboard screenshots.
5. Validation evidence.
6. Executive summary containing key findings and recommendations.

Recommended tools: Power BI, Power Query, DAX, Excel, and SQL.

TRACK B: DATA SCIENCE

Assignment Title

Healthcare Operations: Statistical Investigation & Insight Discovery

Objective

Investigate a healthcare operations problem using exploratory data analysis, statistical reasoning, and evidence-based interpretation.

Task 1: Define the research question

Select one primary question supported by the available data.

Possible questions include:

- How do encounter costs vary across encounter classes?
- Are encounter durations associated with differences in recorded claim costs?
- How do encounter volumes change across years or quarters?
- Do procedure costs differ substantially across frequently performed procedures?
- What characteristics are associated with recorded encounter costs, subject to data availability and appropriate interpretation?

State your research question, its relevance, and the variables required to investigate it.

Task 2: Conduct exploratory data analysis

1. Inspect the relevant dataset or linked datasets.
2. Examine distributions, missingness, potential outliers, and data consistency.
3. Investigate relevant relationships between variables.
4. Create appropriate visualisations.
5. Explain what the results show and what they cannot establish.

Task 3: Form and test hypotheses

Develop at least two hypotheses relevant to your chosen question.

For each hypothesis:

- State the null and alternative hypotheses where a formal statistical test is appropriate.
- Identify the variables and analysis method.
- Explain the assumptions of the chosen method.
- Conduct the analysis.
- Interpret the results in plain language.
- Explain the practical significance and limitations of the findings.

Do not use a statistical test simply because it is available. Choose a method that fits the question, data type, and assumptions.

Task 4: Investigate limitations

Explain how missing values, unequal group sizes, repeated records, potential confounding factors, or limited data coverage could affect your conclusions.

Distinguish between association and causation.

Task 5: Produce an analytical report

Your report must include:

- Research question and objectives.
- Data preparation approach.
- Exploratory findings.
- Hypotheses and statistical methods.
- Results and interpretation.
- Limitations and recommendations.

Required Deliverables

1. Reproducible Python notebook.
2. Data-quality summary.
3. At least five relevant visualisations.
4. Statistical analysis report.
5. Findings and recommendations.
6. Instructions for reproducing the analysis.

Recommended tools: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, and Jupyter Notebook.

TRACK C: MACHINE LEARNING ENGINEERING

Assignment Title

Healthcare Operations: Predictive Model Development & Evaluation

Objective

Design, train, evaluate, and document one predictive machine learning solution that addresses a feasible healthcare operations question.

This is an engineering assignment. A successful submission must demonstrate more than training a model: it must show that the model is evaluated appropriately, reproducible, and suitable for its stated purpose.

Task 1: Select a prediction problem

Choose one feasible target based on the available fields and sufficient valid records.

Potential directions include:

- Predicting encounter duration or whether an encounter exceeds a defined duration threshold.
- Estimating encounter costs using available encounter information.
- Predicting a clearly defined repeat-encounter outcome, if the data supports reliable construction of the target.

You must justify the selected problem and confirm that the target can be constructed from the supplied data.

Do not assume that the available data supports every prediction problem.

Task 2: Define the target and features

Document:

- The target variable and how it is constructed.
- The intended prediction point.
- The features that would be available at that point.
- Features excluded because they would introduce leakage or are unavailable at prediction time.
- Missing-data handling and feature transformations.
- Potential limitations in the target definition.

For repeat-encounter prediction, define the time window and observation period carefully. Do not treat every subsequent encounter as a clinical readmission.

Task 3: Establish a baseline

1. Create a simple baseline model.
2. Select appropriate evaluation metrics.
3. Split the data using a method suited to the problem.
4. Where time-based prediction is intended, consider a chronological split.
5. Explain why the selected validation strategy is appropriate.

Task 4: Train and compare models

Train at least two suitable model approaches, including the baseline, and compare their results.

For classification, consider suitable metrics such as precision, recall, F1-score, and ROC-AUC where appropriate.

For regression, consider MAE, RMSE, and an appropriate baseline comparison.

Explain why the selected metrics matter for the use case.

Task 5: Test and package the solution

- Build a reproducible training workflow.
- Evaluate model errors and limitations.
- Check for data leakage and inappropriate use of features.
- Include basic input validation and tests.
- Document how the model can be loaded and used for inference.
- Explain what further monitoring would be needed before operational deployment.

Required Deliverables

1. Source code and reproducible notebook or scripts.
2. Data preparation and feature engineering documentation.
3. Baseline and model comparison report.

4. Evaluation metrics and error analysis.
5. Saved model where appropriate.
6. Inference script or working prototype.
7. Tests and README.

Recommended tools: Python, Pandas, Scikit-learn, Matplotlib, Git, and an appropriate model-persistence method.

Responsible-use restriction: This task concerns healthcare operations. Do not present the model as a diagnostic or treatment system, and do not claim clinical validity based solely on these datasets.

TRACK D: GENERATIVE AI

Assignment Title

Healthcare Operations Knowledge Assistant

Objective

Build a prototype AI assistant that helps users understand approved healthcare operational metrics and dataset definitions while reducing unsupported or misleading responses.

The assistant is intended for operational information and analytics support, not clinical advice.

Task 1: Define the use case

Identify the intended users and at least five operational questions the assistant should answer.

Examples:

- What does the `TOTAL_CLAIM_COST` field represent?
- Which encounter classes are present in the dataset?
- What is the difference between base encounter cost and total claim cost?
- How is encounter duration calculated?
- What information is needed to analyse payer coverage?

Include questions that the assistant should not answer because the supplied sources do not provide sufficient evidence.

Task 2: Prepare the knowledge sources

1. Review `data_dictionary.csv`.
2. Identify the approved documentation and validated findings that can support responses.
3. Prepare the knowledge sources in a structured format.
4. Document gaps in the available information.
5. Avoid inventing hospital policies, clinical guidelines, or operational rules that are not supplied.

Task 3: Build the application

Create a simple working application that:

- Accepts user questions.
- Retrieves or uses approved information.
- Produces understandable responses.
- Provides source references or explanations of the evidence where feasible.
- Handles unsupported questions appropriately.
- Displays errors clearly without exposing credentials or sensitive information.

For questions involving numerical results, use validated calculations or a controlled query layer rather than asking the language model to invent figures.

Task 4: Evaluate the assistant

Create a test set of at least 20 questions covering:

- Correct, directly supported questions.
- Questions requiring numerical calculations.
- Ambiguous questions.
- Questions with insufficient supporting information.
- Incorrect assumptions.
- Requests for unsupported clinical advice.
- Attempts to obtain unnecessary identifying patient information.

Record the expected behaviour, actual response, whether the response was supported, and any corrective action required.

Task 5: Improve reliability

Document how you addressed inaccurate answers, unsupported claims, privacy risks, and failure cases.

Required Deliverables

1. Working AI assistant prototype.
2. Source code and setup instructions.
3. Description of the knowledge sources and workflow.
4. Test questions and expected outcomes.
5. Evaluation report with observed failures.
6. Privacy and responsible-AI risk assessment.
7. Demonstration video or screenshots.

Recommended tools: Python, an approved LLM API, Streamlit or Gradio, and a retrieval or structured-query approach appropriate to the use case.

Security requirement: Never commit API keys or credentials to source control. Do not send identifiable patient data to an external model unless the data handling has been explicitly authorised and the service is approved for that use.

TRACK E: PROJECT MANAGEMENT

Assignment Title

Healthcare Intelligence Project: Planning, Governance & Delivery Control

Objective

Prepare an execution-ready project management plan for delivering a healthcare intelligence solution using the supplied datasets.

Your plan must account for the different technical workstreams, project dependencies, quality requirements, privacy risks, and delivery constraints.

Task 1: Develop the project charter

Prepare a project charter containing:

- Project background and business need.
- Project objectives.
- Scope and exclusions.
- Key deliverables.
- Stakeholders and responsibilities.
- Assumptions and constraints.
- Acceptance criteria.
- Major project risks.

Task 2: Create the work breakdown structure

Develop a structured work breakdown structure covering the major phases, activities, and deliverables required for a healthcare intelligence project.

Include activities for data assessment, analysis or model development, solution testing, documentation, and presentation.

Task 3: Develop the project schedule

Prepare a Gantt chart with:

- Activities and milestones.
- Estimated durations.
- Dependencies.
- Planned start and finish dates.
- Responsible roles.
- Critical-path analysis where the scheduling method supports it.

Clearly identify assumptions used to estimate durations and dependencies.

Task 4: Prepare the resource and budget plan

Estimate the resources required to deliver the project.

Include appropriate categories such as labour, software, hosting, testing, documentation, and contingency where relevant.

Label figures as estimates and state the assumptions behind them. Do not present hypothetical costs as actual client expenditure.

Task 5: Develop a risk and quality management plan

Prepare a risk register containing at least eight project risks.

Consider:

- Poor data quality.
- Unclear metric definitions.
- Inconsistent source fields.
- Privacy and unauthorised data access.
- Delayed deliverables.
- Model or application failures.
- Inaccurate AI-generated responses.
- Inadequate testing or validation.

For each risk, document its likelihood, impact, mitigation, owner, and status.

Also define quality gates and acceptance criteria for each major deliverable.

Task 6: Simulate project progress

Create a hypothetical progress update in which at least two activities are delayed and one change request is raised.

Explain the impact on dependencies, schedule, cost, scope, and quality. Recommend corrective actions and update the relevant project records.

Clearly label the scenario as a simulation, not an actual client event.

Required Deliverables

1. Project charter.
2. Work breakdown structure.
3. Gantt chart and dependency analysis.
4. Resource and estimated budget plan.
5. Risk register.
6. Quality and acceptance plan.
7. Simulated status report and change log.
8. Final project delivery and handover checklist.

Recommended tools: Microsoft Project, Excel, Jira, Trello, or an equivalent project management tool.

ASSESSMENT CRITERIA

Section	Marks
Business problem understanding	10
Data quality, assumptions, and evidence	15
Technical quality and correctness	25
Testing, validation, and reliability	20
Business interpretation and recommendations	10
Documentation and reproducibility	10
Professional communication	10
Total	**100 Marks**

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 3 project.
- Publish a LinkedIn post describing your project
- Publish an X (Twitter) post summarizing your project.
- Tag AnalystLab Africa on LinkedIn.
- Tag @analystlabafric on X.
- Include the hashtag #AnalystLabAfrica on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.